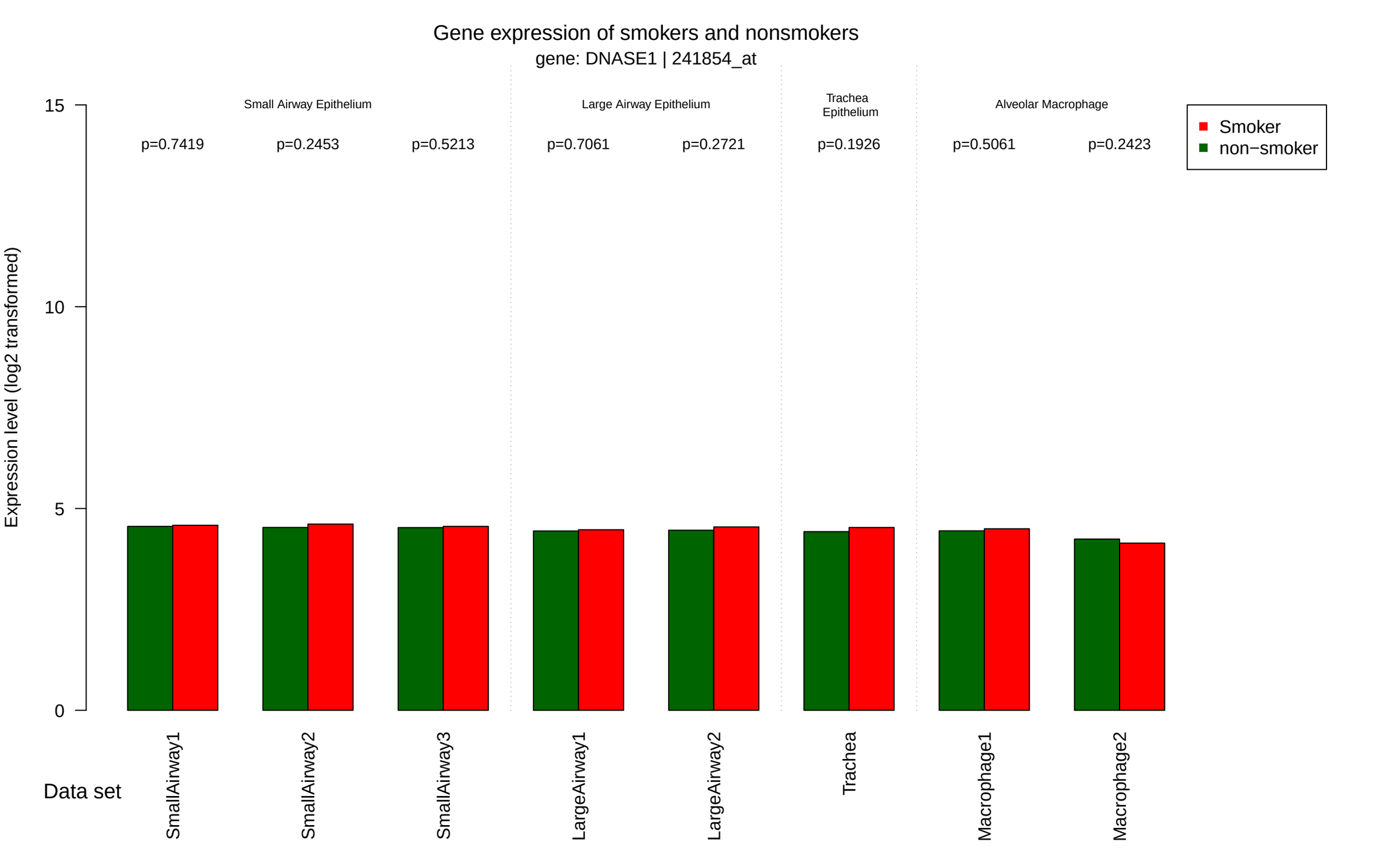
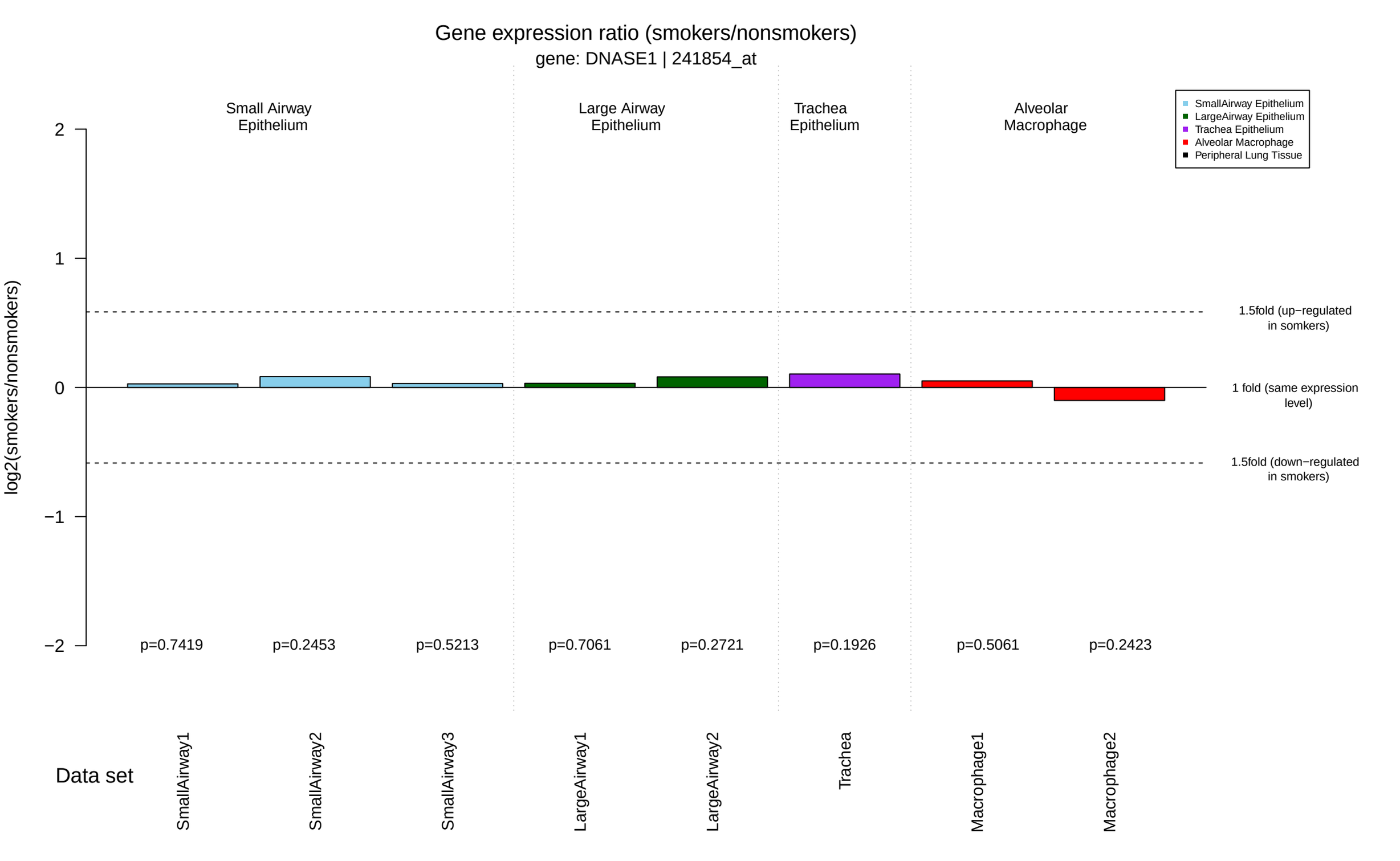
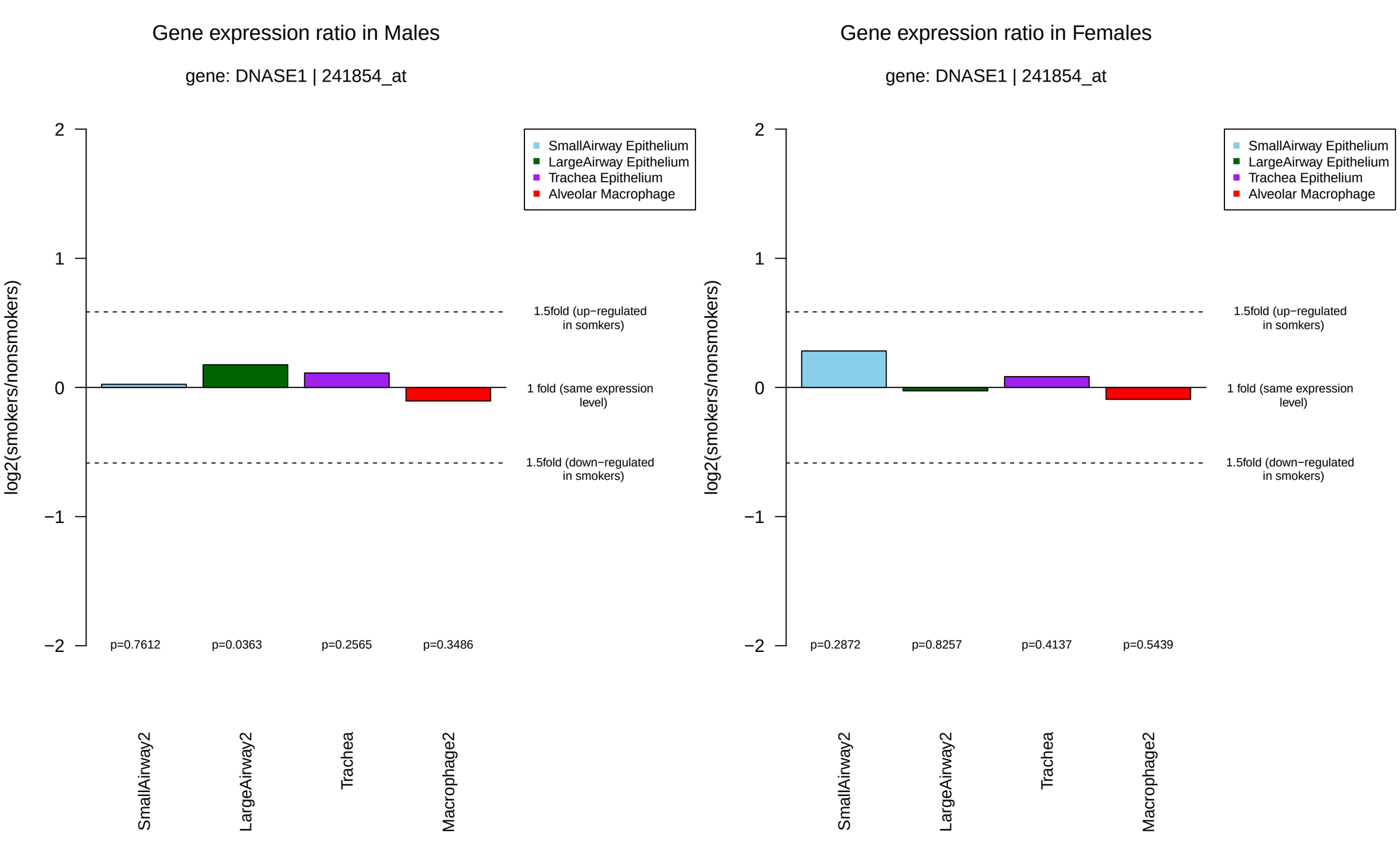
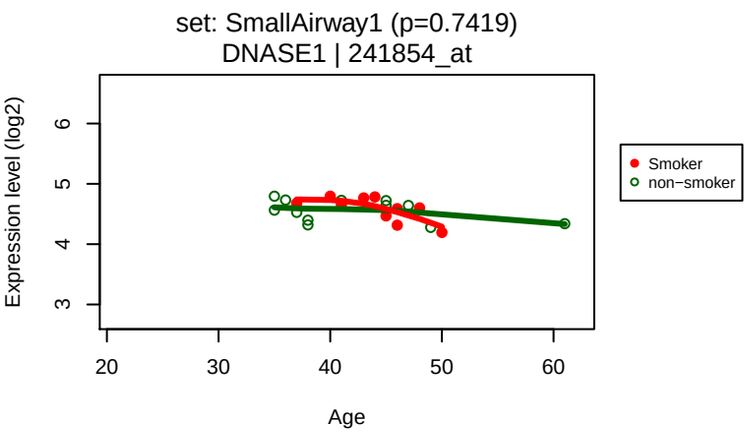
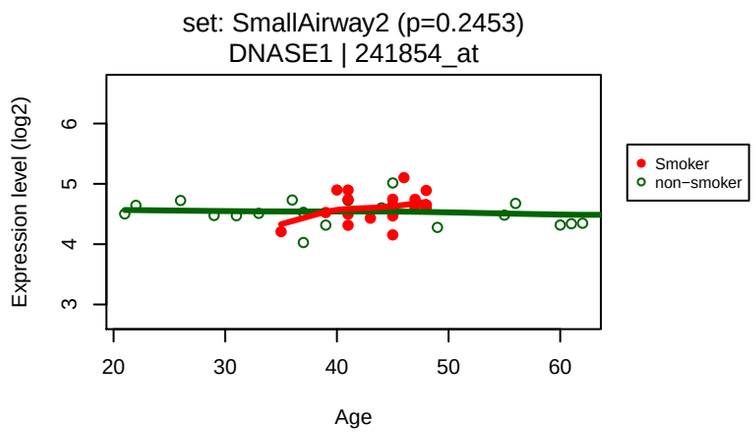
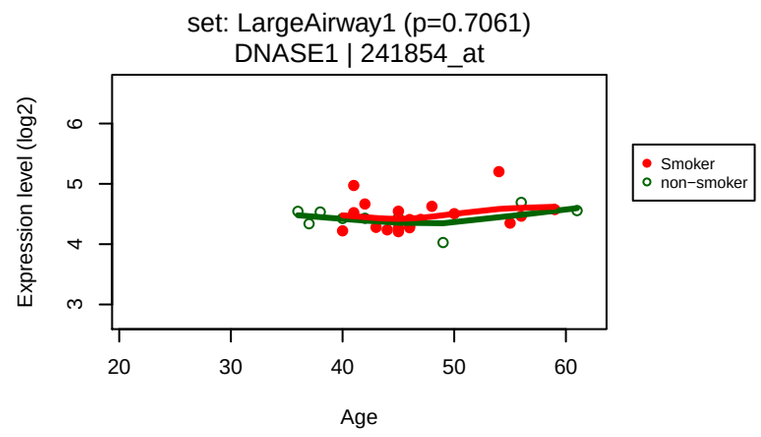

Gene expression of smokers and nonsmokers
gene: DNASE1 | 241854_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7419
p=0.2453
p=0.5213
p=0.7061
p=0.2721
p=0.1926
p=0.5061
p=0.2423
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

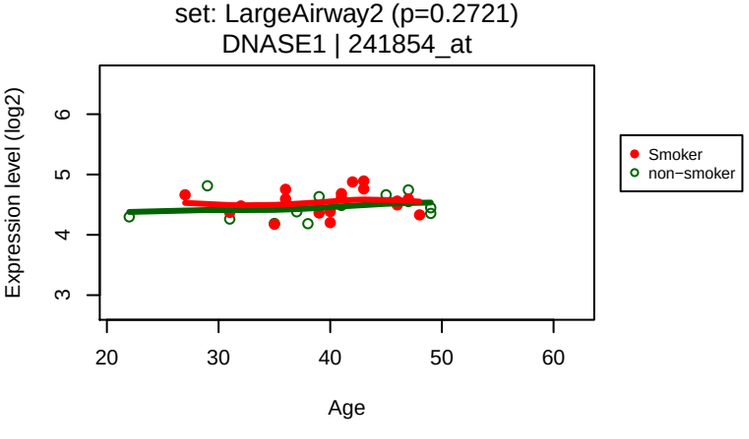

Gene expression ratio (smokers/nonsmokers)
gene: DNASE1 | 241854_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7419
p=0.2453
p=0.5213
p=0.7061
p=0.2721
p=0.1926
p=0.5061
p=0.2423
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

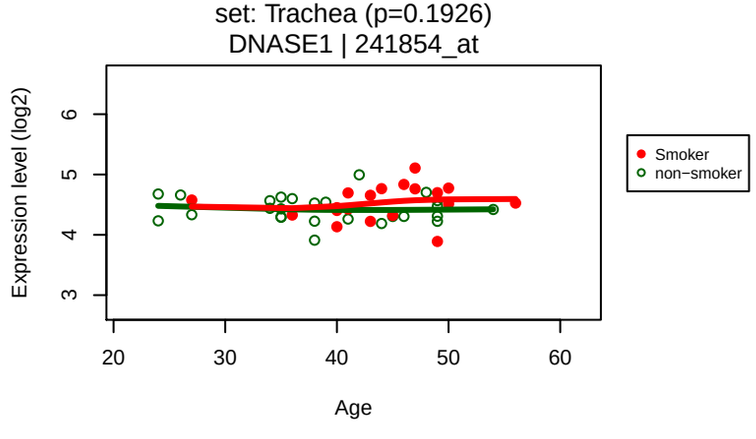

Gene expression ratio in Males
gene: DNASE1 | 241854_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7612
p=0.0363
p=0.2565
p=0.3486
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNASE1 | 241854_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2872
p=0.8257
p=0.4137
p=0.5439
SmallAirway2
LargeAirway2
Trachea
Macrophage2

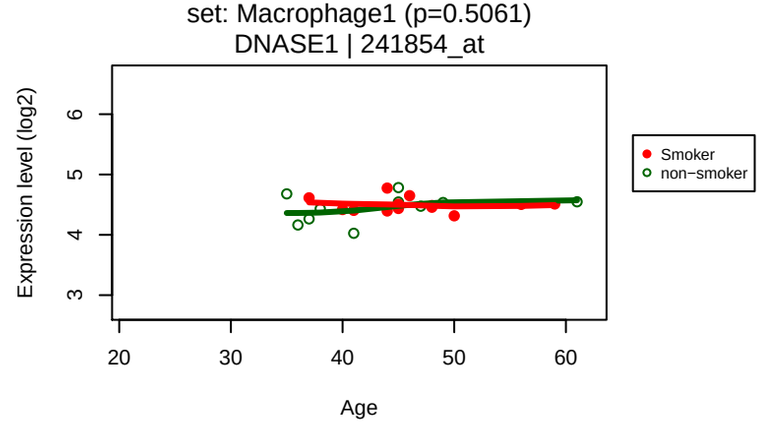

set: SmallAirway1 (p=0.7419)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker

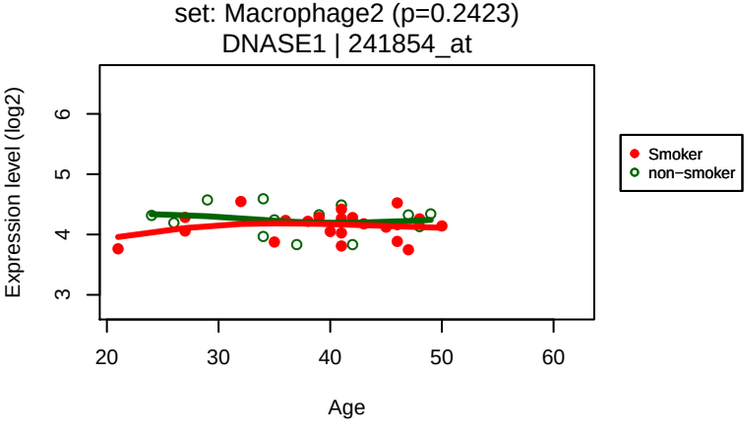

set: SmallAirway2 (p=0.2453)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker

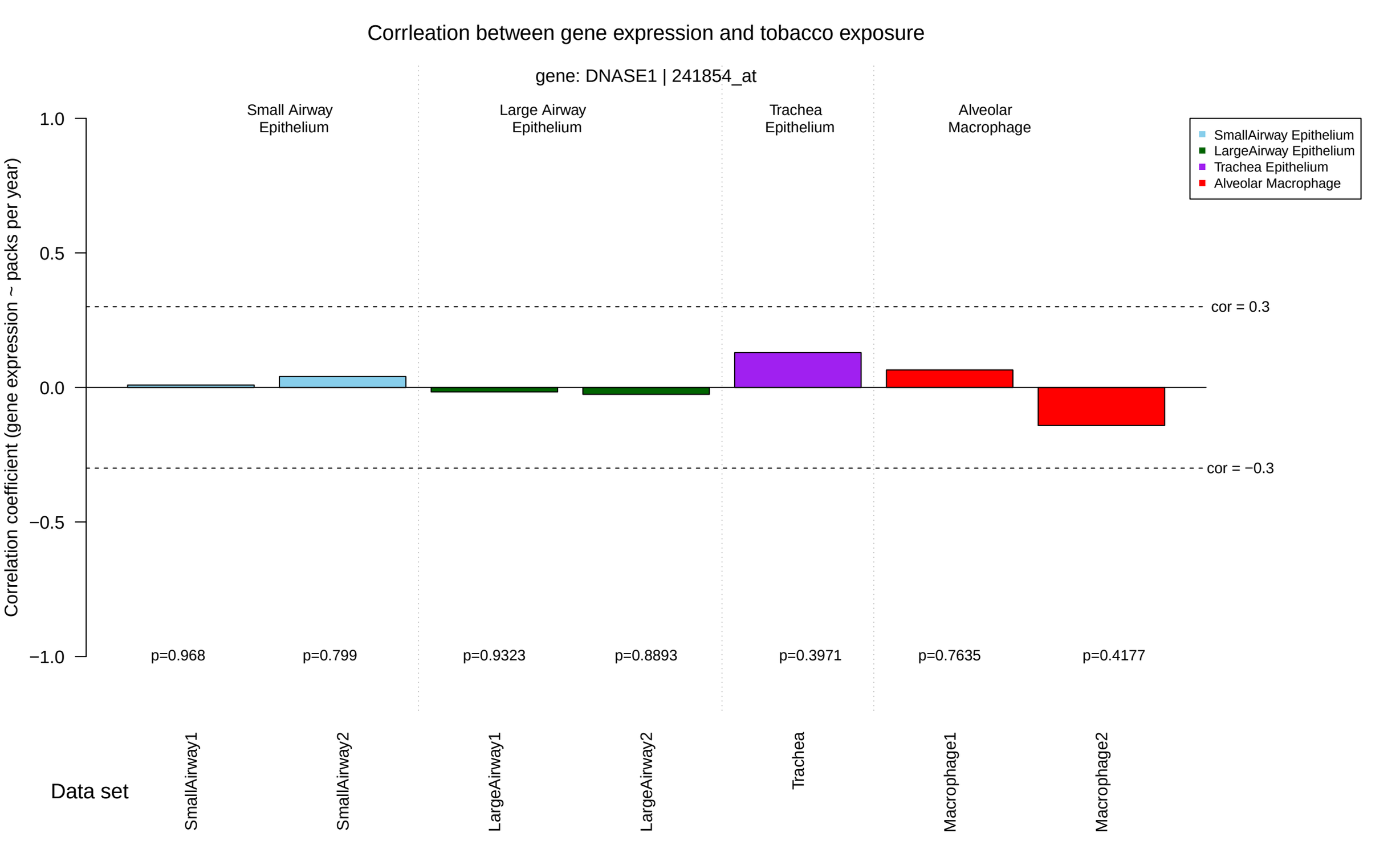

set: LargeAirway1 (p=0.7061)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2721)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1926)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5061)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2423)
DNASE1 | 241854_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNASE1 | 241854_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.968
p=0.799
p=0.9323
p=0.8893
p=0.3971
p=0.7635
p=0.4177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2